Q16 GHS direct-screen isogeny frontier

Exact structural bound

Across all **8,190** normalized ordinary elliptic models over F_{4096}, the triangle, Hess, and nonsplit direct genus-three V4 screens cover **6,158** models. Their entire F_{4096} isogeny closure covers **8,124**. Exactly **66 models in three source-order classes** lie outside that closure. This excludes those models from the three specified direct-cover routes, including all source-field isogeny walks into those routes.

Exact candidate representatives

$K = F_2[w]/(w^{12}+w^7+w^6+w^5+w^3+w+1)$. Each source has $Y^2+XY = X^3+32X^2+b$; all coefficients and coordinates below are polynomial-basis masks.

#E	b	Models	r	G	Q = 7G	LMFDB class
3974	306	12	1987	(307, 2825)	(1678, 342)	3.16.a_a_aet
3978	11	30	17	(2340, 4083)	(2181, 2262)	3.16.a_a_aep
4178	31	24	2089	(3408, 895)	(1008, 2898)	3.16.a_a_dd

Proof and source checks

The exact finite-field replay enumerates each screen and uses a trace-dual Walsh transform to compute every elliptic source order. The 6,158 screened models attain 125 of the 128 orders in the full cohort. The missing orders are 3974, 3978, and 4178, containing 12, 30, and 24 models. The finite-field isogeny theorem makes equal source order equivalent to F_{4096}-isogeny, proving the 8,124/66 split without a walk-radius limit. Sage's native elliptic routine independently recounts the three representatives and verifies their listed subgroup points.

What the exception means

LMFDB already records genus-three hyperelliptic Jacobians in all three classes. The simple-root gate $\gcd(r, 3(1-16^3))=1$ proves that some isogeny preserves each selected prime subgroup. That existence result supplies no evaluated map or attack timing. The order-3978 class was also present in the earlier repository screen; its selected prime has only 17 elements. The order-3974 and order-4178 classes have selected primes 1987 and 2089 and are the stronger map-construction targets.

Complete Magma parameter sweep on representatives

For each exact representative, Magma 2.30-1 exhausted all 4,095 nonzero parameters c in its documented WeilDescentGenus intrinsic: 30 intrinsic rejections, 30 genus-four outputs, 450 genus-six outputs, and 3,585 genus-seven outputs. **Zero parameters gave genus three.** This is a complete parameter-specific result for those equations; it does not exclude other covers or the Jacobian-isogeny route.

Claim boundary and sources

This is a new *finite exclusion certificate for these three screens*, not a new geometric family, new transfer method, or measured GHS advantage. Other covers and correspondences remain possible. Exact equations, all 66 source masks, code, LMFDB labels, and per-candidate prior-work checks are in Q16_FRONTIER.md, frontier_q16.json, and prior_work_frontier_q16.json. Relevant prior methods: Murai et al. 2022 (minimal-genus classification) and Tian 2020/2023 (Weil-restriction isogeny transfer).